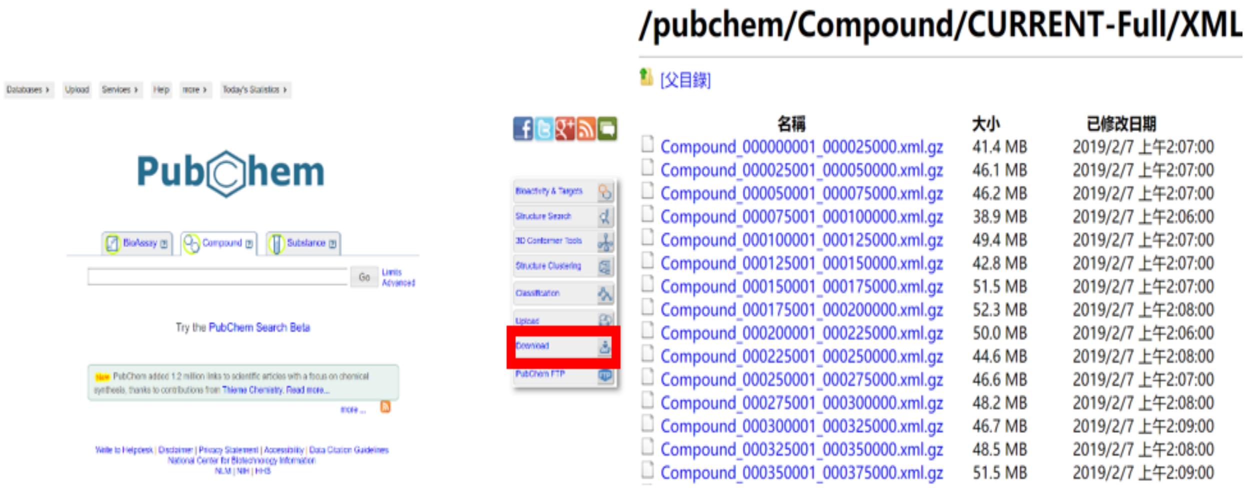
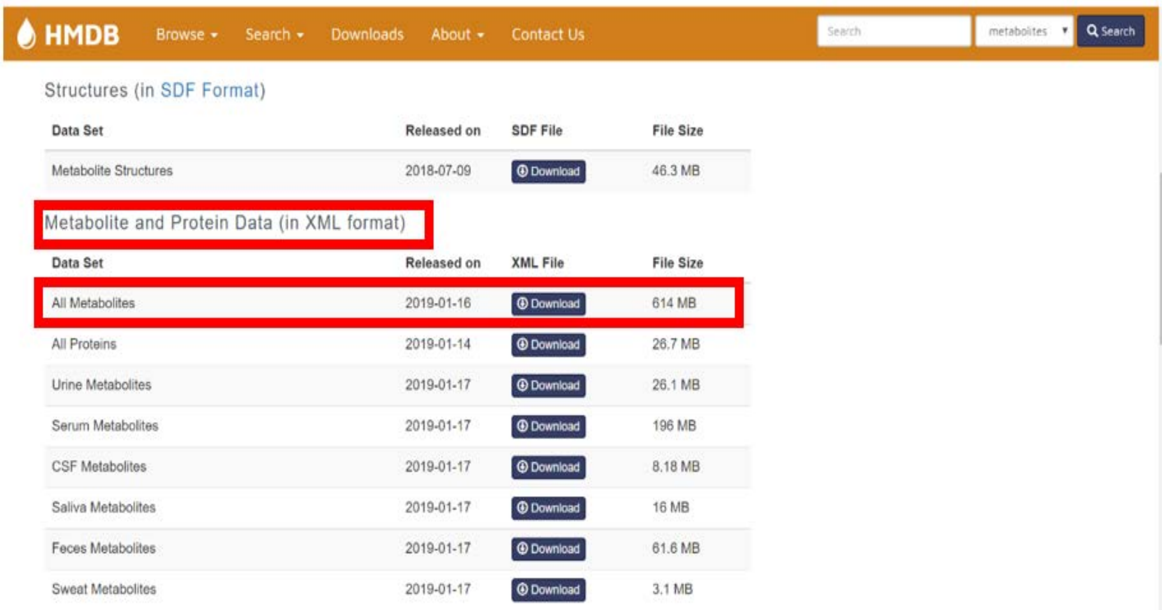
User Guide for Formula Subset Analysis(FSA)

Noted that the datasets excerpted from PubChem and HMDB were downloaded in 2018 and thus the contents in the current web pages now may be different from the snapshots below. Also the file names and the corresponding path may need to be changed in the provided programs. In addition, the provided Python programs are better run under python 3.6 or later and the MATLAB programs are better run under MATLAB 2021b or later.

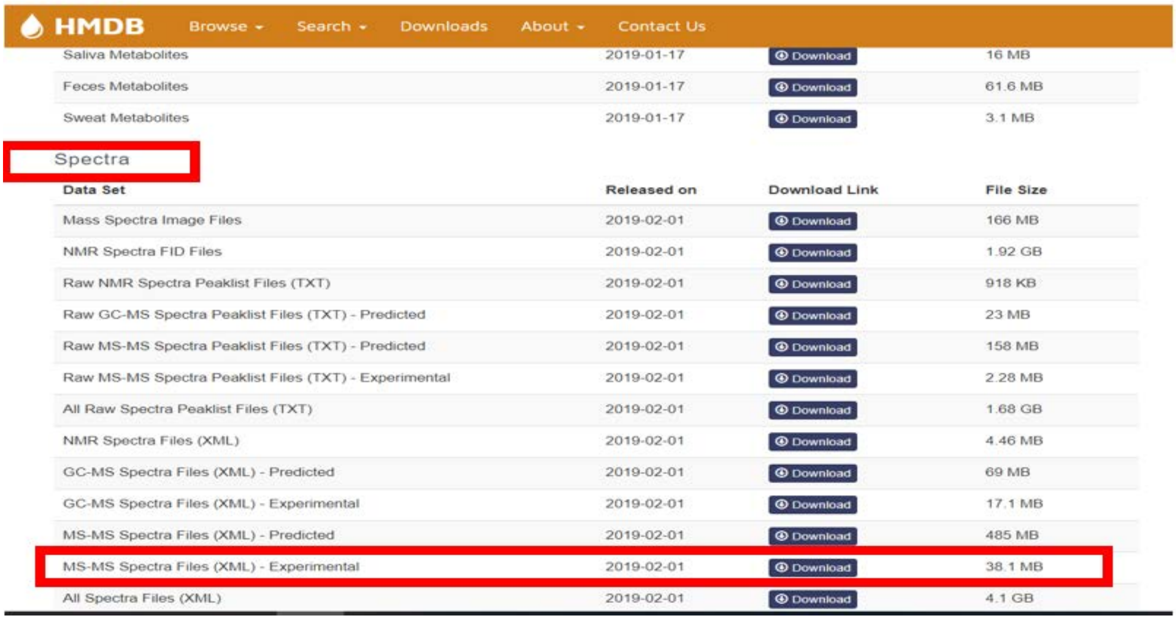
1. Download Data:
   a、 PubChem: For the search of compound mass formula candidates (https://ftp.ncbi.nlm.nih.gov/pubchem/Compound/CURRENT-Full/XML/)



   b、 HMDB Test dataset (http://www.hmdb.ca/downloads)
      i. Compound information



ii. MS/MS spectra



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Data Set	Released on	Download Link	File Size
Saliva Metabolites	2019-01-17	Download	16 MB
Feces Metabolites	2019-01-17	Download	61.6 MB
Sweat Metabolites	2019-01-17	Download	3.1 MB

Spectra

Data Set	Released on	Download Link	File Size
Mass Spectra Image Files	2019-02-01	Download	166 MB
NMR Spectra FID Files	2019-02-01	Download	1.92 GB
Raw NMR Spectra Peaklist Files (TXT)	2019-02-01	Download	918 KB
Raw GC-MS Spectra Peaklist Files (TXT) - Predicted	2019-02-01	Download	23 MB
Raw MS-MS Spectra Peaklist Files (TXT) - Predicted	2019-02-01	Download	158 MB
Raw MS-MS Spectra Peaklist Files (TXT) - Experimental	2019-02-01	Download	2.28 MB
All Raw Spectra Peaklist Files (TXT)	2019-02-01	Download	1.68 GB
NMR Spectra Files (XML)	2019-02-01	Download	4.46 MB
GC-MS Spectra Files (XML) - Predicted	2019-02-01	Download	69 MB
GC-MS Spectra Files (XML) - Experimental	2019-02-01	Download	17.1 MB
MS-MS Spectra Files (XML) - Predicted	2019-02-01	Download	485 MB
MS-MS Spectra Files (XML) - Experimental	2019-02-01	Download	38.1 MB
All Spectra Files (XML)	2019-02-01	Download	4.1 GB

c、 MassBank: The test dataset was used in "Hydrogen rearrangement rules: computational MS/MS fragmentation and structure elucidation using MS-FINDER software". Analytical Chemistry 88, 7946-7958, 2016. The data can be downloaded from the "Table S5" of the paper at https://pubs.acs.org/doi/abs/10.1021/acs.analchem.6b00770.

d、 CASMI2016:

i. Download Challenge Solutions Category 2+3 at http://www.casmi-contest.org/2016/solutions-cat2+3.shtml

ii. Manually add the formula of the compounds to the last column.

iii. Download the peak list file Challenge_negative_peaklist.zip for negative mode and Challenge_positive_peaklist.zip for positive mode at http://www.casmi-contest.org/2016/challenges-cat2+3.shtml.

e、 LipidBlast:

i. Go to the website at https://fiehnlab.ucdavis.edu/projects/lipidblast and click on the "LINK" below.

Download LipidBlast:

The LipidBlast libraries and all LipidBlast material are available for free commercial and non-commercial use under a Creative-Commons By-Attribution (CC-BY) license. By downloading, you agree to correctly attribute the works and libraries. Certain provided materials are excluded from the CC-BY and marked accordingly. You can check the smaller handbook first [PDF] to see if LipidBlast is right for you. Download the supplement data and project software here (direct download).

1. Download LipidBlast (full version, all templates, all spectra, software and handbook, 103 MB) - [LINK] LipidBlast-Full-Release-3.zip (You need to click the download button, please use the free 7ZIP for unpacking) (SHA-256: 930182af12dff7ea74d546f6116410d58cfbb25842bcb7bd8f465e8311781342)

ii. Under the folder LipidBlast-Full-Release-3>LipidBlast-ASCII-spectra, download "LipidBlast-neg.msp" and " LipidBlast-pos.msp".

2. Use **MySQL Workbench** to construct a **compound** database
   a、 Install **MySQL Workbench** and create a user "root" with password "1234" (MySQL Workbench Installation: A Step-by-Step Guide https://www.simplilearn.com/tutorials/mysql-tutorial/mysql-workbench-installation).
   b、 Add a "**compound**" database
      i. Right-click on the list of existing **Schemas** and select **Create Schema.**
      ii. Enter "**compound**" for the schema and for collation choose 'utf - utf8_bin'. Then click **Apply**.
      iii. In the **Apply SQL Script to Database** window, click **Apply** to run the SQL command that creates the schema.
      iv. Click **Finish**. You can see the new schema, which has no tables, listed in the left pane.
      v. From the **Management** menu, select **Users and Privileges** and click **Add Account**. Complete the screen with the credentials listed above. Navigate to the tab **Schema Privileges** and click on **Add Entry...**. Select the newly created database schema. Click **Select "All"** to grant all privileges on this schema for this new user. Click **Apply**.
      vi. From the **Management** menu, select **Options File** and click the **Networking** tab. Find the `max_allowed_packet` entry (should be at the top) and change it to at least `256M`. Click **Apply...** and in the new dialog that appears again **Apply**.
      vii. If you get the error *Could not save configuration file*, you did not run the MySQL Workbench as an administrator. Restart it as an administrator and try again.
      viii. Finally, to apply this change you need to restart the database. From the **Instance** menu, select **Startup/Shutdown** and click **Stop Server**, followed by **Start Server**.
3. Construct PubChem Dataset for Compound Formula Candidates Search
   a、 Use the provided readgzip_COMPOUNDtoMYSQL.py to convert PubChem XML files into MySQL database table "**compounddata20190529**".
   b、 Be sure to adjust the user name and password of the SQL server and the file names and paths of the downloaded files if different names are used.
   c、 Use the following query to extract the qualified records and save them into "PubChem_metabolites.csv"

```
SELECT MolecularFormula, Weight_MonoIsotopic
FROM compounddata20190529 as t1
WHERE t1.Weight_MonoIsotopic < 1500 and (t1.MolecularFormula like '%C%' or
t1.MolecularFormula like '%H%' or t1.MolecularFormula like '%O%' or
t1.MolecularFormula like '%N%' or t1.MolecularFormula like '%S%' or
t1.MolecularFormula like '%P%') and t1.MolecularFormula not like '%F%' and
t1.MolecularFormula not like '%Cl%' and t1.MolecularFormula not like '%Br%' and
t1.MolecularFormula not like '%I%'
INTO OUTFILE 'C:/ProgramData/MySQL/MySQL Server
```

5.6/Uploads/PubChem_metabolites.csv'
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\r\n';

d、 Use Matlab program "convert_PubChem_metabollites.m" to further ensure the element contains CHONSP only and to convert the formula into a numerical matrix indicating number of CHONSP elements in each formula. The "PubChem_metabolites.csv" is converted into output file "PubChemMetabolite.mat" which is need in all the FSA tests in step 5.

4. Construct test datasets

a、 HMDB

i. Use the provided python file HMDBtoMySQL4_0218.py to convert HMDB Compound XML file into MySQL database table "**hmdbdata20190226**".

ii. Use the provided python files MSMStoMySQL.py and MSMStoMySQL_otherproblem.py to convert HMDB MS/MS XML file into MySQL database table "msmsdata20190226"

iii. Use the follow query to extract the qualified records

```
select t1.name, t1.accession, t1.chemical_formula, t1.monisotopic_molecular_weight,
   t2.ionization_mode, t2.instrument_type, t2.collision_energy_voltage, t2.peak_counter,
   t2.ms2_peak
from hmdbdata20190226 as t1
inner join msmsdata20190226 as t2
on t2.database_id=t1.accession
where (t2.instrument_type like '%LC-ESI-QTOF%' or t2.instrument_type like '%LC-ESI-
   ITFT%' or t2.instrument_type like '%LC-ESI-QFT%') and t2.peak_counter > 2 and
   t1.chemical_formula like '%C%' and t1.chemical_formula like '%H%' and
   t1.chemical_formula not like '%F%' and t1.chemical_formula not like '%Cl%' and
   t1.chemical_formula not like '%Br%' and t1.chemical_formula not like '%I%' and
   t1.chemical_formula not like'%B%';
```

iv. Use the "Export" function in MySQL Workbench to save the resultant record to an Excel Spreadsheet xml file "FSA_HMDB_result_CHONSP.xml"

v. Open the file with MS Excel and save it as "FSA_HMDB_result_CHONSP.xlsx".

b、 MassBank:

i. Open the file downloaded from the "Table S5" of the paper at https://pubs.acs.org/doi/abs/10.1021/acs.analchem.6b00770.

ii. Remove compounds with elements other than CHONSP.

iii. Remove precursor types other than [M+H]+ and [M-H]-.

iv. Save the resultant file as ' MSFINDER_MassBank_test_CHONSP.xlsx'

c、 CASMI2016:

i. Remove compounds with elements other than CHONSP.

ii. Remove precursor types other than [M+H]+ and [M-H]-.
iii. Save the resultant file as ' solutions_casmi2016_cat2and3_CHONSP.csv''
iv. Create a folder "CASMI2016_Cat2and3_All\positive" and unzip the files in "Challenge_positive_peaklist.zip" to this folder.
v. Create a folder "CASMI2016_Cat2and3_All\negative" and unzip the files in "Challenge_negative_peaklist.zip" to this folder.

d、 LipidBlast:

i. Under the downloaded fold LipidBlast-Full-Release-3\LipidBlast-ASCII-spectra\, find LipidBlast-neg.msp. Use the provided Matlab program "LipidBlast_MS2_to_Table.m" to convert the file into an MS Excel file "LipidBlast_negative.xlsx". Be sure to change the source and destination file names to LipidBlast-neg.msp and LipidBlast_negative.xlsx in the program.
ii. Under the downloaded fold LipidBlast-Full-Release-3\LipidBlast-ASCII-spectra\, find LipidBlast-pos.msp. Use the provided Matlab program "LipidBlast_MS2_to_Table.m" to convert the file into an MS Excel file "LipidBlast_positive.xlsx". Be sure to change the source and destination file names to LipidBlast-pos.msp and LipidBlast_positive.xlsx in the program.
iii. Use MS Excel to merge the two files LipidBlast_negative.xlsx and LipidBlast_positive.xlsx.
iv. Remove compounds with elements other than CHONSP.
v. Save the resultant file as ' LipidBlast_all_CHONSP.xlsx''

5. Run FSA on the test dataset in Matlab (2021a or later version)

a、 FSA_HMDB.m (for the HMDB dataset): Users need the previous generated "PubChemMetabolite.mat" and "FSA_HMDB_result_CHONSP.xlsx" to run the program. Users can download the provided FSA_HMDB_result_CHONSP.xlsx and PubChemMetabolite.mat since they are authorized to be distributed here.

b、 FSA_Massbank.m (for the MassBank dataset): Users need the previous generated "PubChemMetabolite.mat" and "MSFINDER_MassBank_test_CHONSP.xlsx" to run the program. Users can download the provided PubChemMetabolite.mat but you have to generate the MSFINDER_MassBank_test_CHONSP.xlsx following the previous instructions.

c、 FSM_CASMI2016.m (for the CSSMI2016 dataset) : Users need the previous generated "PubChemMetabolite.mat" and "solutions_casmi2016_cat2and3_CHONSP.csv" and the peak lists in the two folders "CASMI2016_Cat2and3_All\positive" and "CASMI2016_Cat2and3_All\negative" to run the program. Users can download the provided PubChemMetabolite.mat but you have to generate the solutions_casmi2016_cat2and3_CHONSP.csv and the two peak list folders following the previous instructions.

d、 FSA_LipidBlast.m (for the LapidBlast dataset): Users need the previous generated "PubChemMetabolite.mat" and "LipidBlast_all_CHONSP.xlsx" to run the program. Users can download the provided PubChemMetabolite.mat but you have to generate the LipidBlast_all_CHONSP.xlsx following the previous instructions.